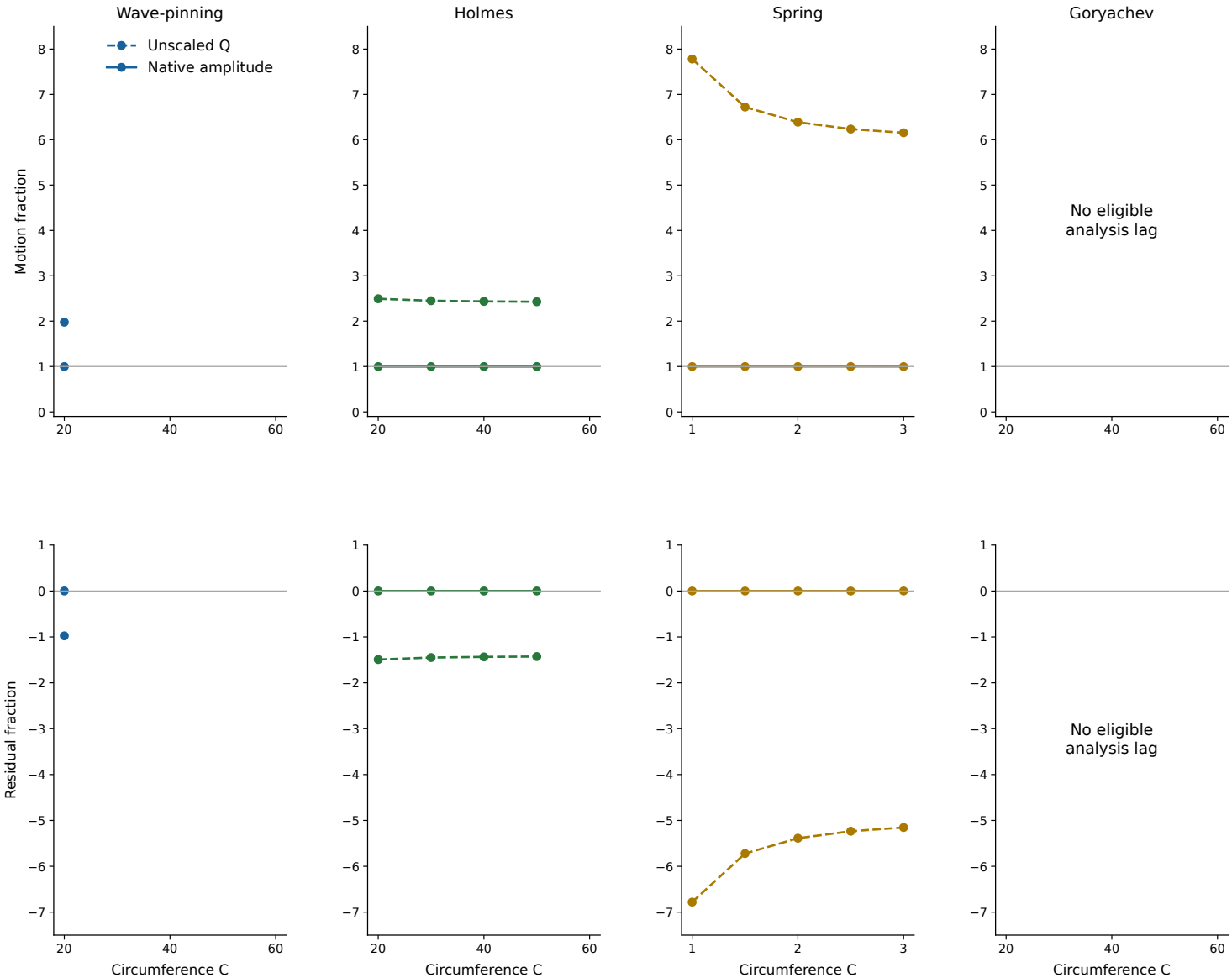
S4.A Native amplitude removes the point-region normalization artefact

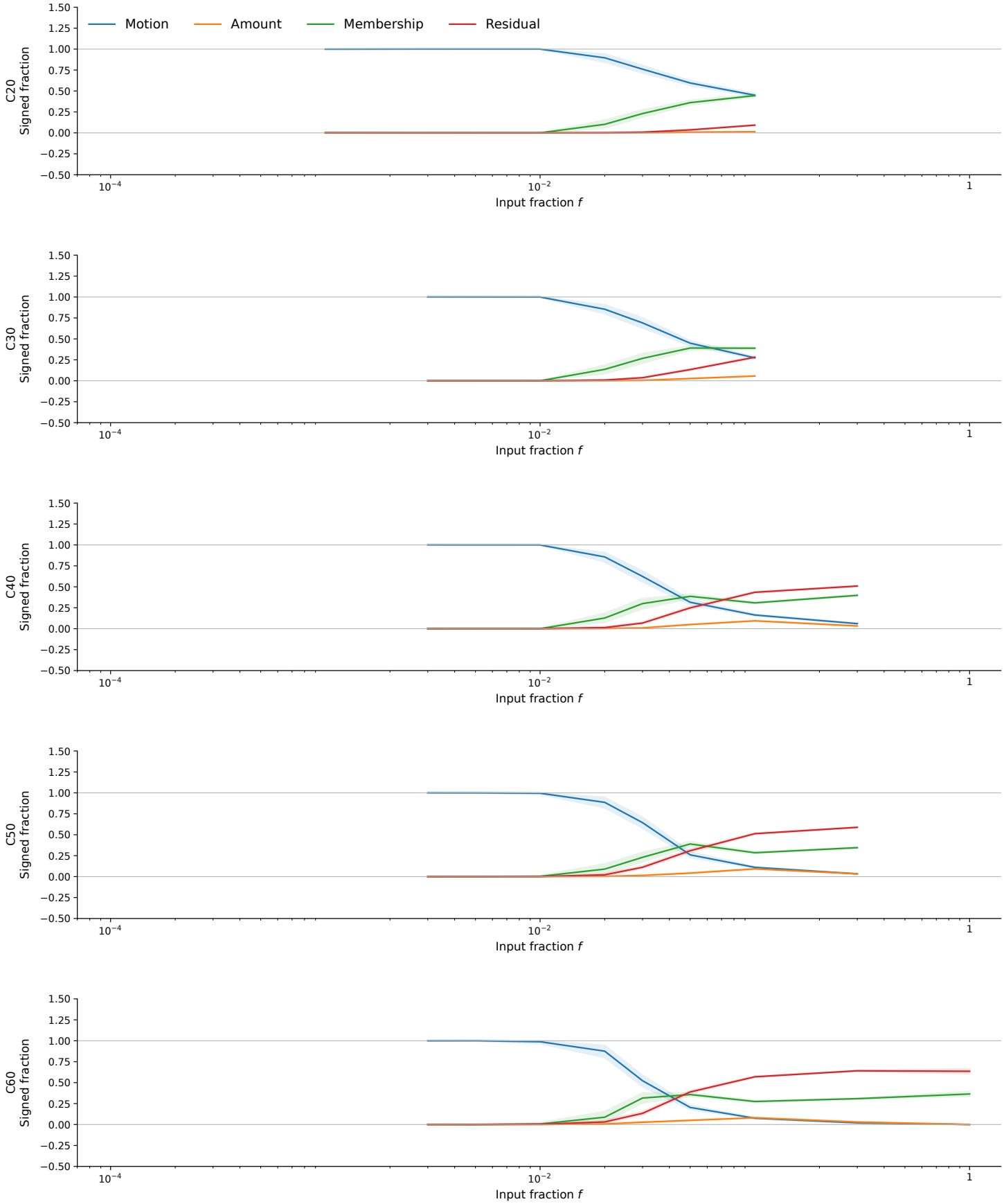


$$Q = \frac{\sum_i A_i e^{i\theta_i}}{\sum_i A_i}, \quad r_t = |P_t|, \quad \Delta P_j = \bar{r}\,\Delta Q_j \ (j = \mathrm{M}, \mathrm{A}, \mathrm{B}) \tag{S4.1}$$

$$\Delta P_{\mathrm{R}} = \Delta P - \bar{r}\,\Delta Q, \qquad \bar{r} = (r_0 + r_1)/2 \tag{S4.2}$$

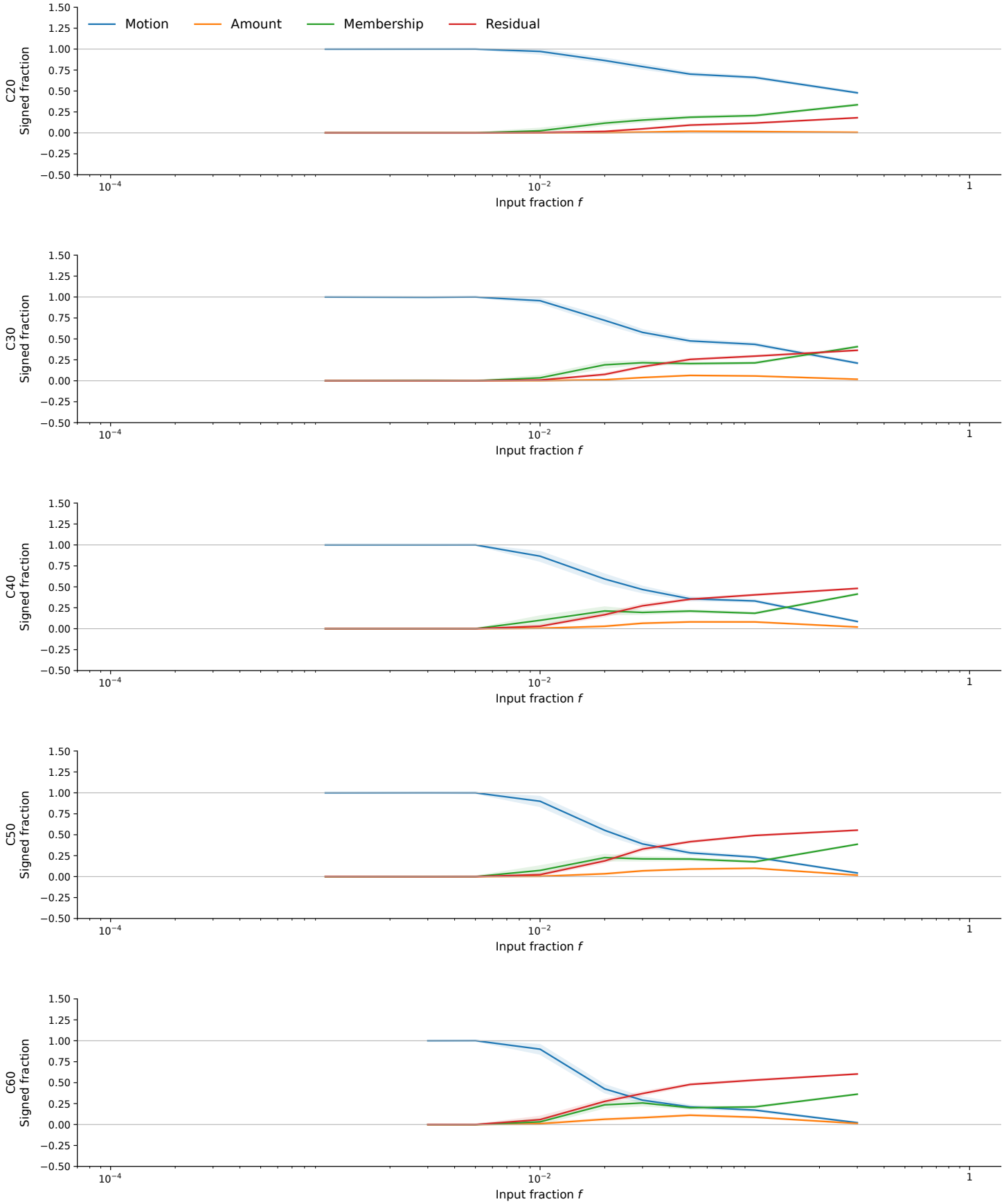
At f=0.001, the original unit-modulus single-region Q creates M≈1/|P| and R≈1−1/|P|.
The main allocation scales each Shapley ΔQ component by mean native amplitude. Pure rigid translation has M=1 and R=0 even with variable amplitude. The residual retains amplitude/shape/background effects. This remains an exact descriptive convention, not causal attribution. Missing Goryachev estimates remain absent.

S4.B Wave-pinning: amplitude-aware signed heading-loss allocation



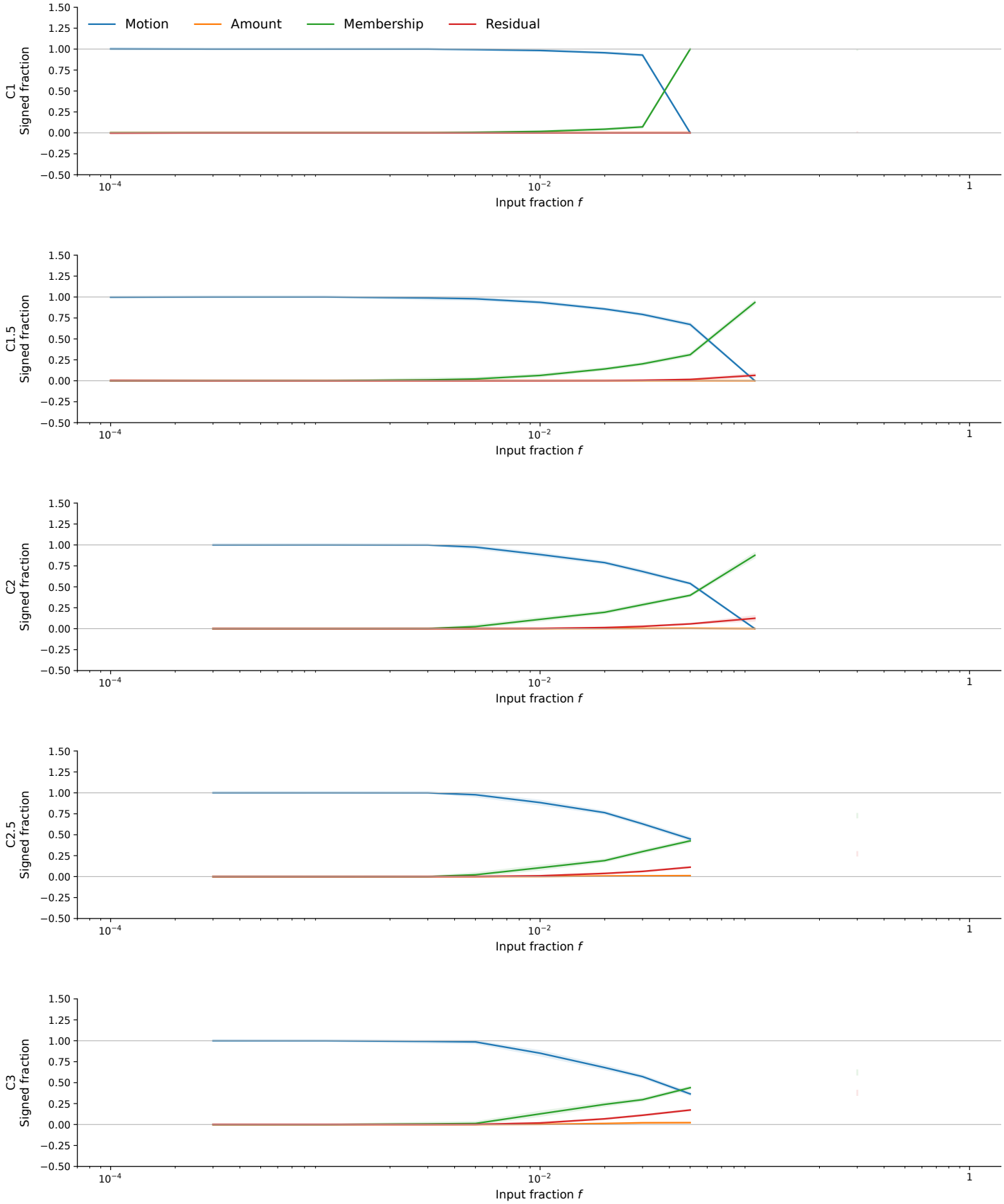
Pointwise whole-history bootstrap intervals (2,000 draws). Source CSV retains values outside the displayed limits.
Same endpoint pairs, lag and raw moments as the unscaled diagnostic; only the accounting convention changes.

S4.B Holmes: amplitude-aware signed heading-loss allocation



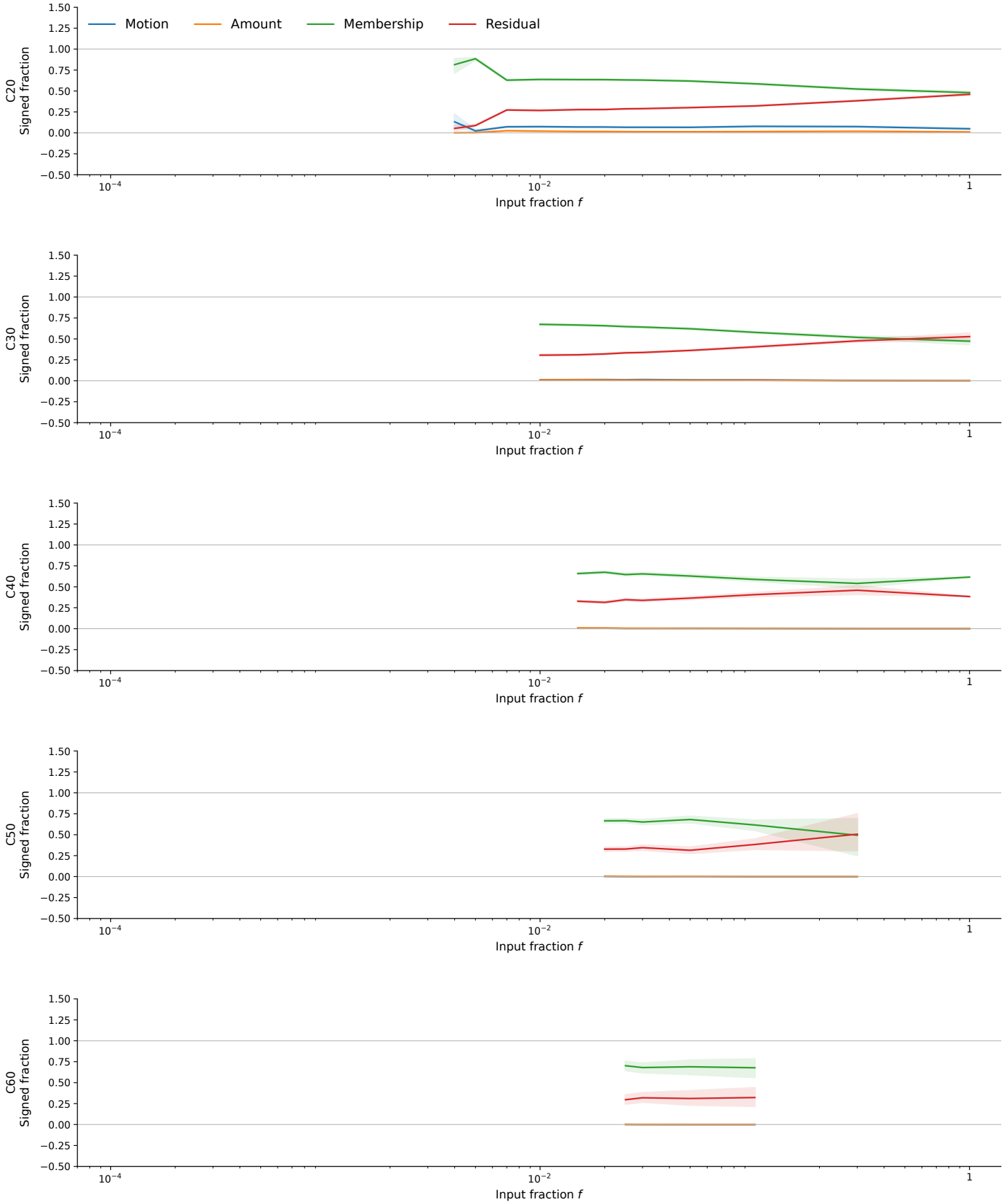
Pointwise whole-history bootstrap intervals (2,000 draws). Source CSV retains values outside the displayed limits.
Same endpoint pairs, lag and raw moments as the unscaled diagnostic; only the accounting convention changes.

S4.B Spring: amplitude-aware signed heading-loss allocation



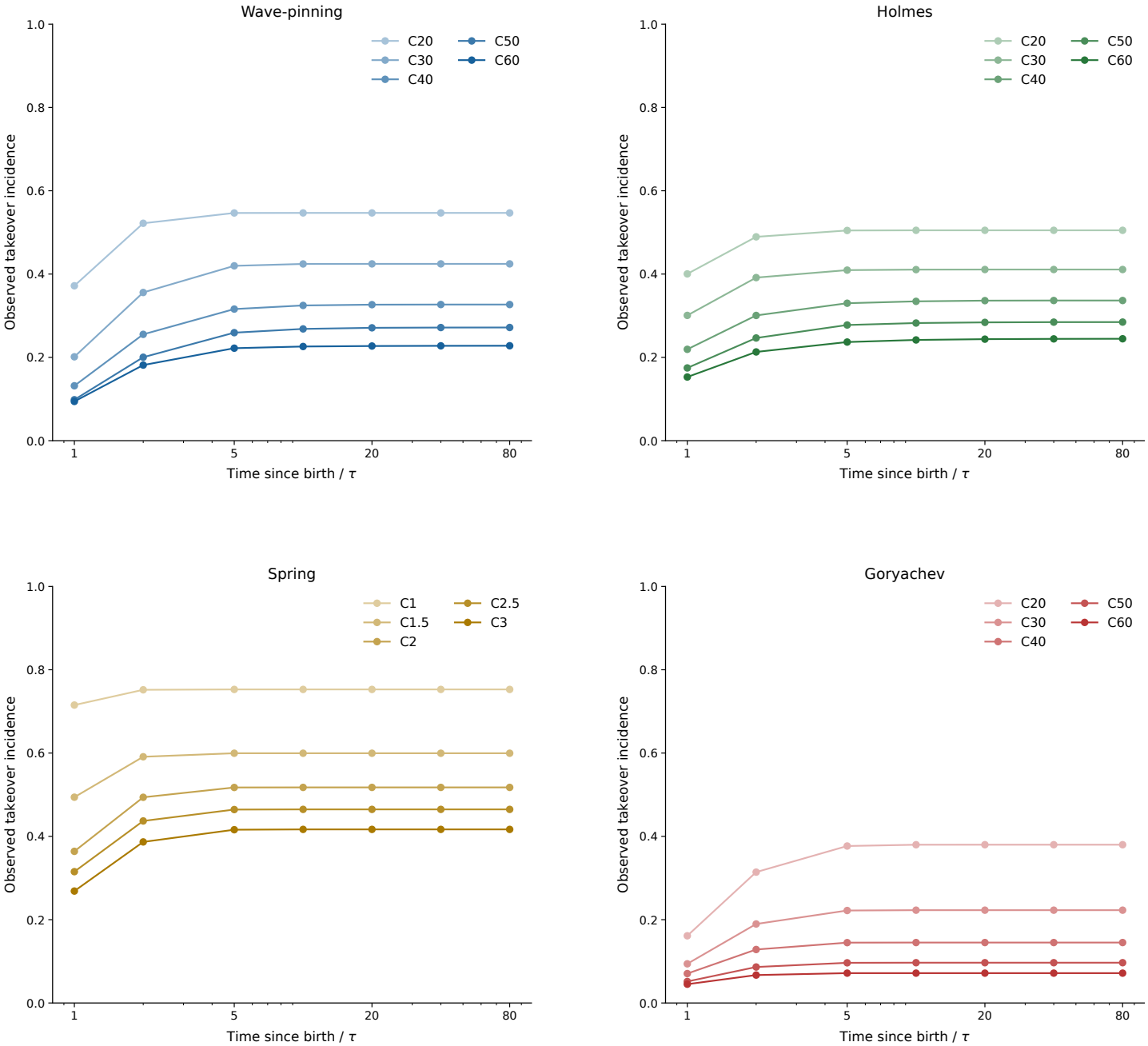
Pointwise whole-history bootstrap intervals (2,000 draws). Source CSV retains values outside the displayed limits.
Same endpoint pairs, lag and raw moments as the unscaled diagnostic; only the accounting convention changes.

S4.B Goryachev: amplitude-aware signed heading-loss allocation



Pointwise whole-history bootstrap intervals (2,000 draws). Source CSV retains values outside the displayed limits.
Same endpoint pairs, lag and raw moments as the unscaled diagnostic; only the accounting convention changes.

S4.C A fixed follow-up cohort separates observation window from outcome



Certified births with at least 80τ available record follow-up; the same cohort is used at every plotted horizon.
All distance bins are included, and f>0.01 is pooled. Curves are descriptive; no extrapolation to lifetime success.
A lower incidence at 20τ can reflect delayed takeover, a lower eventual incidence, or both.